Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 4.1 · Two secantes

The function f is defined on $\mathbb{R}$ by $f(x) = x^2 - 1$. Note A the point of its curve abscisses 1.

1. Calculate the guiding coefficient of the joining splitter A at the point of the curve of abscisse 2.

2. Resume the calculation with the point of the curve of abscisse $1{,}1$.

3. After study of the derived number, determine $f\prime(1)$ and compare this value to the two calculated rates.
   Explain what changes when the second point approaches A.

Exercise 4.2 · One tangent

The function is considered $f(x) = 2x^2 + 1$, defined on $\mathbb{R}$. The objective is to determine the equation of its tangent at the point of abscissa -1, starting from the rate of change.

1. For $h \neq 0$, write and then simplify $\dfrac{f(-1+h) - f(-1)}{h}$.

2. Deduct from it $f\prime(-1)$ Calculate also $f(-1)$.

3. Write down the reduced tangent equation and verify that it passes through the point of contact.

Fiscal year 4.3 · Marginal cost

A company uses the model $C(q) = q^2 + 20q + 100$, where $q \geq 0$ means the quantity produced and $C(q)$ The model extends the quantity by a real variable in order to study small variations in production.

1. Calculate $C(10)$ and $C(11)$, then the rate of change of C between 10 and 11. Interpret this rate in the context.
2. From the rate between 10 and 10 + h, determine $C\prime(10)$.
3. Compare the two results. Explain why the cost of an additional unit is not exactly equal to the derived number in this model.

Exercise 4.4 · Read a tangent

This exercise has two independent situations. A tangent to a curve is a straight: a other point of this line does not necessarily belong to the curve.

1. In the figure of the section on tangent, read the coordinates of two points on the right green. Deduce its slope and check the value announced in the legend.
2. In another situation, the tangent at the curve of a function f at point $A(2; 3)$ goes through $B(4; 2)$. Determine $f(2)$ and $f\prime(2)$ Is it possible to deduce from it $f(4)$?
3. Use this tangent to provide an approximate value of $f(2,(1)$ Make the sign appear " about equal" in the conclusion.

Exercise 4.5 · Controlling an approximation

We want to approach 0 $,98^2$ without first performing the multiplication. curve of the square function at the point of abscisse 1.

1. Write the equation of this tangent and deduce the desired approximation.
2. Then calculate the exact value of 0 $,98^2$. The error is defined as "exact value minus value approach": calculate this error and specify its sign.
3. Explain the result from the identity (1 + $h)^2 = 1 + 2h + h^2$.

Exercise 4.6 · Tangente and intersection

The function f is defined on $\mathbb{R}$ by $f(x) = x^3$. It is admitted that $f\prime(1) = 3$. A student states that tangente and a curve always have one common point.

1. Determine the tangent equation at the curve of f at the point of abscisse 1.
2. Check by developing that $x^3 - 3x + 2 = (x - 1)^2(x + 2)$.
3. Determine all points of intersection of the curve and this tangent. Is it correct?

Exercise 4.7 · Numerical secants

To study the square function in the vicinity of 1, we define, for $h \neq 0$,

$$T(h) = \frac{(1+h)^2 - 1}{h}.$$

The six values 0 shall be used. ,1, 0,01, 0,001 and their opposites.

1. Write a Python function that calculates $T(h)$, then produce a table of six results.

2. Conjecture the value of the number derived from the square function in 1.

3. Simplify algebraically $T(h)$ to justify the conjecture. Explain why the value test
It is not enough to demonstrate the result alone.

Fiscal year 4.8 · Deferred review

The reciprocal function is defined by $f(x) = \frac{1}{x}$ for $x \neq 0$. It is studied in the vicinity of 1.

1. Specify the prohibited values of h in rate $\frac{f(1+h) - f(1)}{h}$, then simplify it.

2. Deduct from it $f\prime(1)$ and the tangent equation in 1.

3. Use this tangent to approach $\frac{1}{1{,}02}$. Maintain the distinction between exact value and value
Come here.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, Exercise 17 p. 127, all questions. Read two points of each tangent; write the
the number of increments rather than estimate its inclination.

CQFD, Exercise 19 p. 127, Questions 1 and 2. Distinguish from point of contact and
slope of the tangent right.